

GENERAL SPECIFICATION G4.25

Tank Scraper Systems

CONTENTS

G4.25	Tank Scraper Systems.....	1
G4.25.1	Submittals.....	1
G4.25.2	Manufacturer’s Qualifications.....	2
G4.25.3	General Design Specification for All Tank Scraper System Configurations	2
G4.25.4	Tank Scraper System with Fixed Bridge and Rotating Bottom Sludge Scraper	5
G4.25.5	Tank Scraper System with Fixed Bridge and Chain-and-Flight Bottom Sludge Scraper ...	5
G4.25.6	Chain-and-Flight Surface Scum Skimmer	7
G4.25.7	Floating Screw Surface Scum Skimmer.....	7
G4.25.8	Reciprocating Surface Scum Skimmer	8
G4.25.9	Miscellaneous Components	9
G4.25.10	Testing	12

G4.25 TANK SCRAPER SYSTEMS

- (a) This specification cover the general requirement of the performance, design and construction of Tank Scraper System equipment in used water (wastewater) application with all relevant statutory regulations and the latest edition of all relevant international, harmonised European and British Standard and Singapore standard and code of practices.
 - (i) The requirements described herein apply to the following types of sedimentation tank scraper system configurations:
 - 1) Fixed bridge sedimentation tank with rotating bottom sludge scraper; and
 - 2) Fixed bridge sedimentation tank with chain-and-flight bottom sludge scraper.
 - (ii) For each of the above configuration types, the surface scum skimmer may be of the chain-and-flight type, floating screw type or the reciprocating type.
 - (iii) Surface skimmer system requirements as described herein are also applicable for implementation in Aeration Grit Tanks.
 - (iv) This specification shall also comply with:
 - 1) The relevant particular specification;
 - 2) The relevant Electrical specification; and
 - 3) The relevant Instrumentation Control and Automation (ICA) specification.
- (b) The configuration type of tank scraper system with rotating bridge over the sedimentation or clarifier tank shall not be used.
- (c) Rotating surface skimmer design for removing floatables also shall not be used.
- (d) Where the process unit of the gravity sludge thickener raking system (picket fence type scraper) is required, the Contractor shall refer to the General Specification G4.55 Sludge Consolidation Thickeners.

G4.25.1 Submittals

- (a) The Contractor shall provide complete fabrication and assembly drawings, together with detailed specifications and data covering materials, power drive, assemblies, and accessories forming a part of the equipment supplied. The data and specifications for each unit shall include, but shall not be limited to, the following:
 - (i) Outline and general arrangement drawing of the equipment indicating the designation overall dimension/foot print and typical layout of key components/system;
 - (ii) Overall dimensions and estimated weights of all furnished equipment;
 - (iii) Component drawings showing all details of construction;
 - (iv) Comprehensive listing of materials of constructions of parts and components guaranteed for durability and against corrosion;

- (v) Weight schedule for each major equipment component and the heaviest individual maintenance and erection lifts;
- (vi) A typical control philosophy;
- (vii) A typical commissioning plan (including timescale for completion);
- (viii) A lubrication schedule detailing parts requiring lubrication, and method and type and manufacturer of lubricants;
- (ix) Identification, size, rating and location of all instrument and electrical connections;
- (x) The Energy consumption and electrical drive loadings list;
- (xi) Wiring diagrams which show internal wiring, external connection wiring and interconnection wiring requirements. Manufacturer shall indicate the power supply and control power supply requirements;
- (xii) List of all instruments and controls, along with function; and
- (xiii) Equipment test certificates.

G4.25.2 Manufacturer's Qualifications

- (a) The materials covered by these specifications are intended to be standard equipment of proven reliability and as manufactured by a reputable manufacturer having experience in the production of such equipment. The equipment furnished shall be designed, constructed, and installed in accordance with the best practices and methods and shall operate satisfactorily when installed as shown in the Drawings and operated per Manufacturer's recommendations.
- (b) The manufacturer shall have manufactured at least twenty (20) comparable units which have been used for similar service each of which has been operating satisfactory for a minimum of four (4) years.
- (c) The manufacturer shall submit an installation list with contact information for the equipment quoted verifying to the S.O. that the specified manufacturer's qualifications are met. The list shall have information of the installed systems' performances, sizes, capacities and noteworthy features.

G4.25.3 General Design Specification for All Tank Scraper System Configurations

- (a) The performance, design, construction and testing of the equipment shall comply with all relevant statutory regulations and the latest edition of all relevant international and British Standards.
- (b) The equipment and associated plant and instrumentation shall be suitable for the site location and environment and shall fit for the operation condition.
- (c) The equipment shall be designed to collect sludge from the tank floor and scum from the tank surface and transfer it to a hopper or channel for disposal unless specified otherwise.
- (d) The equipment shall be designed for continuous automatic operation in an exposed environment. Maintenance requirements shall be minimised using sealed-for-life bearings. Where lubrication is required lubrication points with automatic lubricator shall be readily accessible.

- (e) The equipment shall provide permanent safe access for operation and maintenance. Where tanks are elevated, stairs or access ladders from ground level shall be provided.
- (f) The weight of the individual components over 25 kg, that are likely to be removed for maintenance shall either:
 - (i) Incorporate clearly identified, permanent lifting points located to give a safe balanced lift;
 - (ii) Be designed in such a way that standard lifting accessories (e.g. slings) can easily be attached and sling slippage is prevented; and
 - (iii) With respected to item (i) above, if lifting points are not designed for lifting the complete component, they shall be clearly marked accordingly.
- (g) Materials
 - (i) The material of the plant and equipment and its associated components shall be suitable for the site location and operation environment. Corrosion protection system shall be provided if required and necessary and shall submit to S.O. for approval;
 - (ii) All fasteners, anchor bolts, nuts and washer shall be of 316 stainless steel; and
 - (iii) All sealant used shall be suitable for the purpose.
- (h) Support Structure
 - (i) The structure shall be constructed in steel. All welds shall be continuous;
 - (ii) All metallic items (such as fixing and fixtures etc.), which are continually or routinely wetted or submerged or immersed in by wastewater shall be made of stainless steel 316L unless otherwise specified;
 - (iii) Support structure that is never submerged shall be of galvanised mild steel material. Alternatively, these structure shall be protected against corrosion by the application of paints or protective coatings such as glass flake or fibres epoxy coating, etc. subject to the S.O.'s approval;
 - (iv) All applied galvanising shall comply with BS EN ISO 1461 required and necessary;
 - (v) The height from the top of the tank wall to the bridge walkway shall not exceed 450 mm;
 - (vi) Support structure shall be of a robust design and be adequately braced to ensure rigidity under all operating conditions;
 - (vii) The bridge structure shall have two parallel beams of steel channel section, positioned to present a flat surface at the outside edge of the bridge;
 - (viii) For all the structural calculations including the design of handrails and floor plates shall be design and endorsed by P.E. and submitted to S.O. for approval;
 - (ix) Support frames shall be fabricated to withstand equipment and material loads;
 - (x) All metal fabrications (i.e. frames, support structure etc.) shall be designed to prevent the collection or trapping of liquids and debris and, if appropriate, facilitate the application of paint system and protective coatings; and

- (xi) The end of all steel sections shall be sealed, to prevent ingress of liquids, after corrosion resistant coatings have been applied.
- (i) Scraper Bridges
 - (i) The steel bridge shall be designed in accordance with the relevant parts of BS EN 1993 (Eurocode3). Bridge support beams shall be channels, with their flanges facing inwards or I section beams. Unless otherwise specified, the bridge support beams shall be continuous (i.e. no joints positioned at mid-span or locations of high stress/moments). The structure shall be designed and endorsed by P.E. and the endorsed details of the designs shall be submitted to S.O.'s approval.
 - (ii) The bridge structure shall not be constructed from any type of hollow steel section. Structural steel section thickness shall not be less than 8 mm.
 - (iii) Bridges shall be designed to support a combined loading made up of the following components:
 - 1) Loads associated with self-weight, walkway flooring, the sludge and scum collecting mechanism's, sludge loading on the blades, raise/lower drive, bridge travel drive, pump and pipework shall be taken into design calculation; and
 - 2) Maximum vertical deflection of beams shall be no more than 1/360th of span.
 - (iv) The structure shall absorb the reactive torque of the complete drive unit installed on the bridge together with a maximum loading created when the tank is either full or empty. The structure shall be able to withstand any imposed loads occurring during start up after a prolonged shut down.
 - (v) Internal cross bracing shall be provided to stiffen the structure and give additional resistance to twisting movements generated from the scraper blades and any drive forces.
- (j) Walkways and Hand-Railings
 - (i) A bridge walkway shall provide access from the tank periphery to the centre of the tank. The walkway shall have a minimum clear width of 600 mm between hand-railings. A full bridge shall have access to the centre of the tank from either end;
 - (ii) All loads including self-load (eg. steel beam members, chequer plate, hand-railing, etc.), scraper drive unit with gearbox and scraper assembly, all attached fixtures and paraphernalia as well as maximum number of maintenance personnel required on the bridge walkway shall be accounted for in designing its load bearing capacity;
 - (iii) The walkway and hand-railing shall comply with General Specification G1.4 Metalwork;
 - (iv) The flooring shall be galvanised mild steel chequer plate forming a continuous walkway for access to carry out routine servicing and maintenance;
 - (v) Flooring shall be sectional with each floor section individually secured to the bridge main beams;

- (vi) Removable sections panel shall be provided as necessary to allow access to equipment assembly below the walkway for inspection and maintenance;
- (vii) The maximum deflection of an unfixed flooring section shall be 1/200th of its longest length;
- (viii) Guide rail shall comply with BS EN ISO 14122-3 or equivalent;
- (ix) Access ladders shall be completely integrated with the hand-railing on each side. The ladder shall be permanently fixed to the bridge and shall overhang the tank wall. Each ladder shall be with safety-cage having lockable swing gate at the point of entry to prohibit unauthorised entry; and
- (x) All bridge mounted equipment shall be located within the main bridge structural members at or above walkway level to allow ready and safe access from the walkway to all items of plant requiring maintenance. Provision for lifting appurtenances shall be made as necessary to facilitate all envisaged lifting operations required.

G4.25.4 Tank Scraper System with Fixed Bridge and Rotating Bottom Sludge Scraper

This configuration type of tank scraper system shall be as follows:

- (a) Material of construction for the scraper assembly and drive shaft shall be SS316L.
- (b) Scrapers shall operate with blade tip speeds in the range of 10 to 90 mm/s so as not to disturb the settlement of solids in the tanks. Selection of the tip speed shall also be subject to any special characteristics of the sludge and proprietary performance enhancing design features;
- (c) The blade shape shall cause the minimum disturbance to settled sludge when the scraper is rotating;
- (d) The scrapers shall be constructed so that they can be operated when the tank is empty without detrimental effects to the equipment;
- (e) Replaceable wearing strips may be neoprene, rubber or brushes, depending on the application and shall be fitted to the blade to provide a continuous contact surface which shall be adjustable;
- (f) Where rubber is used, backing strips shall be fitted to give support to the fixing of the rubber wearing strips. The backing strips shall not project beyond the tip of the scraper blade and the rubber wearing strips shall not project by more than three times their thickness;
- (g) The assembly shall be secured by means of stainless steel fasteners with PTFE washer as a minimum or equal or better compatible materials.

G4.25.5 Tank Scraper System with Fixed Bridge and Chain-and-Flight Bottom Sludge Scraper

This configuration type of tank scraper system shall be as follows:

- (a) Chain and flight type scrapers shall provide a continuous sweep of the tank floor and shall be able to operate with the tank full or empty.
- (b) Two pairs of endless chains wall mounted on sprockets at low level inside the tank shall carry the buoyant scrapers.

- (c) The drive unit shall include a motor driven speed reducing gearbox comprising primary and secondary gears within a common housing such that lubrication arrangements are common to both sets of gears. The drive unit shall be mounted adjacent to the tank wall and shall have an endless chain arranged to drive the submerged transmission shaft which shall drive both wall mounted chains.
- (d) The primary drive chain, the twin immersed chains carrying the scrapers, and the associated sprockets shall be manufactured from nylon or a similar non-corrosive, wear resistant material.
- (e) Continuously or intermittently submerged metallic parts shall be non-corrosive stainless steel or material suitable for the application.
- (f) There shall also be provision of platforms with all necessary appurtenances to safely and conveniently access, service, carry out maintenance works, install, dismantle and re-assemble parts and components of the drive unit and gearbox;
- (g) Sufficient number of scraper flights shall be arranged in echelon formation to deflect sludge progressively towards the draw off point; and
- (h) Chain and flight type scrapers shall have multiple flights mounted on endless chains. There shall be at least one pair, but the number provided shall depend on the size of the process tank.
- (i) Each scraper shall be arranged to scrape the tank floor and convey the sludge into the sludge hopper at one end of the tank.
- (j) The ends of the tubular boom shall be securely fixed to the scraper chains with stainless steel bolts.
- (k) All required fixtures and fittings shall be provided.
- (l) There shall be two end support carriage assemblies each of which shall house the runner wheels and drive arrangement.
- (m) The carriages shall be fabricated from carbon steel adequately braced to withstand all imposed loads. A continuous skirt and covers shall be provided all-round the carriage to protect against operator injury. Sections of the skirt and covers shall be removable to permit inspection and maintenance.
- (n) The rails shall form the track for the runner wheels. If the drive system is of the “rack and pinion” type, this may be integral with the guide rails or be a separate rail. The rails and/or drive rack shall be designed such that water cannot be trapped. The drive pinion should be readily accessible for replacement and should be designed to wear in preference to the drive rail;
- (o) Rails shall be constructed using durable and corrosion-resistant materials such as polypropylene (PP), fibre-glass reinforced plastic (FRP) or stainless steel SS316 with the necessary load bearing capacity. The selected material shall be subject to the S.O.’s approval;
- (p) Limit switches and end stops shall be fitted to all running rails.

G4.25.6 Chain-and-Flight Surface Scum Skimmer

The chain-and-flight surface scum skimmer may be integrated with any of the above tank scraper system configurations. Its requirements shall be as follows:

- (a) Scum and grease shall be steadily swept across the surface and collected into a wet well, scum trough or scum box depending on the design configuration.
- (b) The chain and flight mechanism shall be mounted to the inner surface of the walls of the tank with the sprocket wheels secured near to the four corners.
- (c) The pair of parallelly aligned chains that are wound round the sprockets shall be attached with the flight between them.
- (d) The scum scraper blade fixed to the flight shall move along with the rotation of the chain mechanism.
- (e) The scraper blade shall be capable of moving floatables steadily into the collecting trough with minimum disturbance that may cause the floatables to break up and even sink into the tank.
- (f) The arrangement shall be designed to ensure the minimum unskimmed dead area at each end of the tank.
- (g) The end support carriages and track rails fitted with limit switches and end stops shall comply to similar requirements as described for the chain-and-flight bottom sludge scraper.
- (h) The scum blade shall be capable of being raised clear of top water level.
- (i) Each end of the scum blade shall be provided with a replaceable side wall scraper.
- (j) The blades shall be fixed using continuously rigid support.
- (k) All blades shall have replaceable reversible abrasion resistant wearing squeegee strips. The length of each squeegee strip shall not exceed a third of the blade length.

G4.25.7 Floating Screw Surface Scum Skimmer

The floating screw surface scum skimmer may be integrated with any of the above tank scraper system configurations. Its requirements shall be as follows:

- (a) The floating screw conveyor scum skimmer system shall, through the rotation of one or more helicoids, move grease and scum towards and into a sip and suction unit.
- (b) When the floating screw conveyor scum skimmer system is in operation, the helicoid blades shall be partially submerged below the water surface level to move grease and scum towards and into the sip and suction unit.
- (c) The floating screw conveyor scum skimmer system shall include but not be limited to the following major components and appurtenances:
 - (i) Floating shafted helicoid or double helicoids for moving scum and other floatables;
 - (ii) Driver motor(s) as required for driving the helicoid(s);

- (iii) Sip and suction unit; and
- (iv) Drive chain, sprockets and associated components where applicable for driving the helicoid(s).
- (d) The minimum design features for the floating screw conveyor scum skimmer system shall be as follows:
 - (i) Adjustable elevations of the floating helicoid(s) and sip and suction unit's hopper;
 - (ii) Require minimal maintenance with only periodic lubricating oil change;
 - (iii) Adaptability to fluctuating water levels;
 - (iv) Adaptability to varying thicknesses and characteristics of grease and scum; and
 - (v) Helicoid(s) drive motor(s) shall be maintenance-free and lubricated for life.

G4.25.8 Reciprocating Surface Scum Skimmer

The reciprocating surface scum skimmer may be integrated with any of the above tank scraper system configuration. Its requirements shall be as follows:

- (a) The skimmer shall consist of scraper blades fixed to sledges that are pulled along by drive bars.
- (b) In the forward movement of the reciprocating motion, the blades shall sweep scum and other floatables into a scum pipe, beach or well.
- (c) The entire motion of the drive bars in the forward direction may be pre-determined to be in steps of smaller distances instead of a continuous movement from start to end of the travel.
- (d) The portion of the blades immersed below the water level shall be adjustable to facilitate the maximum movement of floatables but minimum water removal. Only the blades will be in contact with the water surface. No other parts will contact the water.
- (e) At the end of the forward travel, the scraper blade swings up and the drive bars retract the sledges with the blades in one continuous motion back to the start of the travel.
- (f) The arrangement shall be designed to ensure the minimum unskimmed dead area at each end of the tank.
- (g) The scum blade shall be capable of being raised clear of top water level.
- (h) Where tank configuration permits, each end of the scum blade shall be provided with a replaceable side wall scraper.
- (i) The blades shall be fixed using continuously rigid support.
- (j) All blades shall have replaceable reversible abrasion resistant wearing squeegee strips. The length of each squeegee strip shall not exceed a third of the blade length.

G4.25.9 Miscellaneous Components

(a) Weir Plate

- (i) Weir plate, where specified, shall be 316L stainless steel, minimum thickness 6 mm;
- (ii) The weir plates shall be security fixed to the wall of the tank outlet channel. The fixings (anchor bolts) shall be made of stainless steel and have high pull out load and high shear load characteristics. Fixings shall allow adjustment of the weir level by +/-25mm;
- (iii) The type of the weir plates shall be according to the particular specification where specified; and
- (iv) Twin sealing strips shall be provided between the weir plates and the wall of the tank outlet channel. These shall be sufficiently thick to accommodate minor irregularities in the wall and be made of a material that will not deteriorate due to contact with the used water.

(b) Diffuser Drum / Centre-Well

- (i) The system shall have a diffuser drum or centre-well, supported from the bridge and concentric about the centre of the tank. The diameter of the diffuser drum shall be as stated in the Particular Specification. The depth shall be 1500 mm with 200 mm freeboard;
- (ii) The drum shall be manufactured from glass reinforced plastic (GRP), with wall and flange thicknesses not less than 6 mm. All surfaces shall be smooth and crevice free and all cut edges shall be sealed;
- (iii) Two diametrically opposed cut-outs shall be provided in the diffuser drum wall, complete with scoops to assist continuous scum skimmer from inside the drum. Each cut-out shall be 150 mm minimum width x 75 mm minimum height above and below normal top water level;
- (iv) To ensure a rigid construction the drum shall have stiffening flanges at top and bottom edges, and at intermediate points, if necessary. Diffuser drums with segmented construction shall have externally bolted flanged joints. Sufficient fasteners shall prevent joint separation. All bracings shall be external to the drum. The internal area of the drum shall be clear of any obstructions other than picket fence supports, and these shall be without flat surfaces that would prevent solids settling into the hopper;
- (v) The drum assembly, when in an empty tank, shall withstand wind loads of up to 40 m/s; and
- (vi) A central riser pipe and bend terminating in the diffuser at top water level shall be provided. Pipework feeding the central riser (under tank base) shall be supplied and installed by the civil contractor unless specified otherwise.

(c) Bearings and Lubrication

- (i) Equipment component combinations that are likely to come into contact and be subject to relative motion shall be adequately lubricated or manufactured from materials with self-lubricating properties;

- (ii) Bearings shall conform to the standards of the Anti-Friction Bearing Manufacturers Association, Inc. (AFBMA);
 - (iii) Rolling element bearing shall be rated for a minimum L-10 life expectancy of 50,000 hours while operating at maximum load;
 - (iv) Bearing housings or assemblies shall be provided with seals and self-aligning features, if appropriate, to prevent the ingress of dirt and water and/or accommodate misalignment;
 - (v) Lubrication points on bearing housing containing sealed for life bearing shall be fitted with caps to prevent inadvertent bearing re-grease;
 - (vi) Where sealed for life bearings are not provided, automatic lubricant dispensers shall provide a clear indication of the remaining lubricant available. The minimum capacity/life of the automatic grease dispensers shall be jointly determined by the main equipment system supplier/manufacturer and lubrication system provider under the coordination of the main Contractor;
 - (vii) Lubrication points on bearing housing shall be readily accessible without the need to remove guards or covers. If lubrication points are not readily and safety accessible, they shall be connected via lubrication pipework to readily accessible lubrication points, installed on a common battery plate. The spacing of the lubrication points on the battery plate shall allow the retrofitting of automatic grease dispensers if required;
 - (viii) Automatic lubricant dispensers shall provide clear indication of lubricant contents. The Contractor shall provide the first fill of lubricants to all components/system requiring lubrication. All bearings and lubrication pipework shall be fully charged with grease; and
 - (ix) Fixed bridge scrapers shall include a flanged motor driven helical worm gearbox. The secondary output shaft will be connected to the scraper mechanism. An adjustable overload device shall be incorporated in the secondary gear train to protect the equipment and this shall be complete with a torque limiter and safety cut-out switch for connecting directly to the control circuit of the motor starter.
- (d) Gears and Gear Drives
- (i) Unless otherwise specified, gears shall be of the helical or spiral-bevel type, designed and manufactured in accordance with AGMA Standards, with a minimum service factor (thermal and mechanical) of 1.7, a minimum L 10 bearing life of 100,000 hours and a minimum efficiency of 94 %;
 - (ii) Gears and gear drives as part of an equipment assembly shall be shipped fully assembled for field installation;
 - (iii) Gearboxes and motors shall have an ingress protection rating conforming to BS 5420: IP 55 classification;
 - (iv) Material selections shall comply with AGMA values and the manufacturer's recommendations. Input and output shafts shall be properly designed for the service and load requirements. Gears shall be computer-matched for minimum tolerance variation. The output shall have two positive seals to prevent oil leakage;

- (v) Oil lubricated gearbox shall be fitted with oil filling and drain points and where appropriate, an oil breather and /oil level indicator. Oil level and drain location shall be readily accessible;
- (vi) The oil filling and drain points shall be designed so that oil can be easily drained and replaced without spillage;
- (vii) Where gear drive output shafts connect to couplings or sprockets, the gear drive manufacturer shall supply matching key;
- (viii) The gearbox type, number of stages, manufacturer, and service factor and final drive ratio shall be stated in the submitted data sheet; and
- (ix) All gears shall run in oil, and oil pans shall have means for fillings and draining the oil without dismantling any of the screen components. Bearings:
 - 1) Bearings shall conform to the standards of the Anti-Friction Bearing Manufacturers Association, Inc. (AFBMA); and
 - 2) Except where otherwise indicated, bearings of process equipment shall have a minimum L-10 life expectancy of 100,000 hours.
- (e) Drive Units and Control panel
 - (i) All drive system components (e.g. motors, gearbox, drive shafts etc.) shall be adequately supported and provided with lifting eye-bolts to facilitate lifting and overhauling;
 - (ii) The drive shall be maintainable from either the walkway or from outside the tank. If the drive is maintainable from outside the tank, portable access equipment shall be provided if necessary;
 - (iii) All drive shaft assemblies shall be designed for ease of maintenance;
 - (iv) Protection system shall be provided to protect against jamming and other mechanical overloads. The motor shall be automatically tripped in the event of an overload;
 - (v) All drive components shall be adequately supported so that fatigue or under stress does not occur;
 - (vi) The output of the gearbox to chain-and-flight assemblies shall be transmitted to the drive wheels located in each end carriage via drive shafting. Auto-resetting torque limiting protection devices shall be provided between the gearbox and the drive shafting; and
 - (vii) Gears shall be oil lubricated and the gear housing shall incorporate oil filling point, oil breather, and oil level sight glass, drain plug and drip tray sized to accommodate one oil change.
 - (viii) A local control station shall be provided and shall fully comply with the Electrical and ICA specification;
 - (ix) Loss of motion protection shall be provided and shall be proposed to S.O. for approval;
 - (x) All control devices shall be readily accessible for maintenance. Sensors, limit switches etc;

- (xi) Effective guards shall be fitted to all revolving shafts, couplings, end carriage wheels and sprockets in accordance with PD 5304.

G4.25.10 Testing

- (a) The equipment and plant shall be field tested after the erection and in the presence of the S.O. and manufacturer representative to confirm and verify the following:
 - (i) The structural, components and system are compliance to the specification; and
 - (ii) The installation and assembled are correct and in accordance with the relevant P&ID / GA drawings.
- (b) The field test acceptance test shall include demonstration that the equipment operate continuously without vibration, jamming or overheating and perform its specific function satisfactorily.
- (c) The minimum requirement for the functional test shall be as following:
 - (i) Verification of all electrical control functions;
 - (ii) Confirmation checks that the installation is concentric with the tank structure;
 - (iii) Running checks to determine the correct tracking of the wheel path; and
 - (iv) Checks to determine that the scrapers blades provide a complete sweep of the tank floor and retain their position within the helical pattern.
- (d) Performance Test shall be conducted on each unit as following at minimum:
 - (i) Checks to determine the accuracy of the weir plate levels;
 - (ii) Demonstration of the scum skimmer mechanism operation;
 - (iii) Demonstration of the effluent channel and weir cleaning system (if applicable);
 - (iv) Test for continuous 3-hour period without malfunction;
 - (v) Perform with the presence of S.O.;
 - (vi) Test Log: Upon completion of test, record information listed on sample test log; and
 - (vii) Adjust, realign, or modify units and retest if necessary.

END OF SECTION